

**The University of Nottingham**

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, SPRING SEMESTER 2011-2012

**MACHINE LEARNING**

Time allowed: **Two Hours**

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*Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced*

**Answer All Questions**

*Only silent, self contained calculators with a Single-Line Display or Dual-Line Display are permitted in this examination*

*Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.*

*No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.*

***DO NOT turn your examination paper over until instructed to do so***

**Question 1**

Briefly sketch the respective procedure of building a handwritten digit recognition system using each of the following four machine learning techniques

- (a) Multilayer Perceptron Neural Network (5 marks)
- (b) K-Nearest Neighbours (5 marks)
- (c) Support Vector Machines (7 marks)
- (d) Naïve Bayesian (8 marks)

You can assume training and testing data have already been collected and appropriately processed. Using diagrams and drawings as appropriate.

**Question 2**

Although a single Perceptron can only classify two classes, it is possible to combine several Perceptrons together to classify more than two classes. For example, combining the outputs of two Perceptrons can distinguish four classes, and combining the outputs of three Perceptrons can distinguish eight classes, and so on.

An experiment has generated 2-dimensional feature vectors belonging to four classes as shown in the following table

| 2 Dimensional Feature Vector | Class |
|------------------------------|-------|
| (-4, 6)                      | A     |
| (6, 6)                       | B     |
| (4, -6)                      | D     |
| (-4, -4)                     | C     |
| (-6, 4)                      | A     |
| (4, 4)                       | B     |
| (-6, -6)                     | C     |
| (6, -4)                      | D     |

Using as many Perceptrons as appropriate, design a system that will be able to distinguish the four classes. Your answer should include

- (a) A diagrams of each Perceptron. (5 marks)
- (b) The values of all connection weights of each Perceptron. (5 marks)
- (c) The equation of each Perceptron's decision boundary. (5 marks)
- (d) An explanation of how your design has achieved the goal. (10 marks)

**Question 3**

In an object recognition task, it is known that the objects come from one of two classes,  $C_1$  or  $C_2$ . Each instance of an object  $X$  has four features,  $X = (x_1, x_2, x_3, x_4)$ . An experiment has collected 14 instances and their feature values and classification are shown in the following table

| $X = (x_1, x_2, x_3, x_4)$ |       |       |       | Classification |
|----------------------------|-------|-------|-------|----------------|
| $x_1$                      | $x_2$ | $x_3$ | $x_4$ | $C$            |
| 1                          | 1     | 1     | 1     | $C_1$          |
| 1                          | 1     | 1     | 2     | $C_1$          |
| 2                          | 1     | 1     | 1     | $C_2$          |
| 3                          | 2     | 1     | 1     | $C_2$          |
| 3                          | 3     | 2     | 1     | $C_2$          |
| 3                          | 3     | 2     | 2     | $C_1$          |
| 2                          | 3     | 2     | 2     | $C_2$          |
| 1                          | 2     | 1     | 1     | $C_1$          |
| 1                          | 3     | 2     | 1     | $C_2$          |
| 3                          | 2     | 2     | 1     | $C_2$          |
| 1                          | 2     | 2     | 2     | $C_2$          |
| 2                          | 2     | 1     | 2     | $C_2$          |
| 2                          | 1     | 2     | 1     | $C_2$          |
| 3                          | 2     | 1     | 2     | $C_1$          |
|                            |       |       |       |                |
| 1                          | 3     | 1     | 2     | = ?            |
| 2                          | 2     | 2     | 2     | =?             |
| 3                          | 1     | 1     | 1     | =?             |

Design a decision tree classifier to class unknown feature vectors  $X_1 = (1, 3, 1, 2)$ ,  $X_2 = (2, 2, 2, 2)$  and  $X_3 = (3, 1, 1, 1)$ . You must show the detail of the working process. An answer without the working process will not receive any marks.

Multilayer Perceptrons handle linearly non-separable classes by introducing hidden layers with non-linear activation functions, which allow the network to learn complex non-linear decision boundaries through back-propagation. In contrast, Support Vector Machines address non-linearly separable data by using soft margins and kernel functions to map data into a higher-dimensional space where a linear separator can be found. While MLP training is based on gradient descent and may converge to local minima, SVM training solves a convex optimization problem and guarantees a global optimum with maximum margin. (25 marks)

**Question 4**

- (a) Compare and contrast how Multilayer Perceptron (MLP) and Support Vector Machine tackle linearly non-separable classes.

sol Over-fitting occurs when a model fits the training data too closely and captures noise rather than the underlying pattern. It can be alleviated by controlling model complexity, for example through regularization, feature selection or dimensionality reduction, and by using validation data-based techniques such as cross-validation and early stopping. Increasing the amount of training data can also help improve generalization and reduce over-fitting. (10 marks)

- (b) Discuss the similarity and difference of K-means and K-nearest neighbours (KNN)

(8 marks)

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- (c) Discuss what causes over-fitting in machine learning and why it should be avoided.

Over-fitting can be reduced by limiting model complexity using regularization, selecting appropriate model parameters via cross-validation, and stopping training early to prevent the model from memorising the training data. (7 marks)

Both K-means and K-nearest neighbours use distance measures to evaluate similarity between data points and involve a parameter  $K$ . However, K-means is an unsupervised learning algorithm that clusters unlabeled data by minimising within-cluster variance, whereas KNN is a supervised learning algorithm that classifies new instances using majority voting among the  $K$  nearest labelled samples. K-means builds an explicit model using cluster centroids and has higher training cost but lower testing cost, while KNN stores all training data and has minimal training cost but higher testing cost during prediction.

Over-fitting occurs when a model learns not only the underlying patterns of the training data but also noise.

It is mainly caused by overly complex models with too many parameters, insufficient training data, noisy or irrelevant features, and excessive training. As a result, the model may memorise the training data instead of learning generalisable patterns.

Over-fitting should be avoided because it leads to poor generalization performance.

Although an over-fitted model achieves low training error, it performs poorly on unseen test data, making its predictions unreliable and reducing its practical usefulness.